

Problem 5

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Problem 1 (a) True or False: If f and g are two functions defined on $(-1, 1)$, and if $\lim_{x \rightarrow 0} g(x) = 0$, then it must be true that $\lim_{x \rightarrow 0} (f(x) \cdot g(x)) = 0$.

(b) True or False: If f is continuous on $(-1, 1)$, and if $f(0) = 10$ and $\lim_{x \rightarrow 0} g(x) = 2$, then

$$\lim_{x \rightarrow 0} \frac{f(x)}{g(x)} = 5.$$

(c) True or False: If f is continuous on $[1, 3]$, and if $f(1) = 0$ and $f(3) = 4$, then the equation $f(x) = \pi$ has a solution in $(1, 3)$.

(d) True or False: Let f be a positive function with vertical asymptote $x = 5$. Then

$$\lim_{x \rightarrow 5} f(x) = \infty.$$